

Predictive Modeling of Residential Property Prices in Pierce County

Edberto Moura Lima

Gerta Hajrullahi

Helga Deutschmann

Mira Radakovic

2026-05-11

Table of contents

Preface	4
Project Team	4
Document Status	4
1 Project Objective	5
1.1 Background & Motivation	5
1.2 Project Goal	5
1.2.1 Qualitative Objectives	5
1.3 Quantitative Objectives	6
1.4 Deliverables & Timeline	6
2 Data Source Description & Data Dictionary	7
2.1 Overview	7
2.2 Relational Diagram	7
2.3 Data Dictionary Summary	7
3 Project Organization & Task Allocation	9
3.1 Group Structure	9
3.2 Tools & Infrastructure	9
3.3 Project Kanban Board	9
3.4 Risk Management	10
4 Development Plan Overview	11
4.1 Workflow Diagram	11
4.2 Work Package Summary	12
4.3 Communication & Oversight	12
5 Project Schedule & Work Packages	13
5.1 Interactive Gantt Chart	13
5.2 Detailed Work Package Definitions	13
5.2.1 WP0: Project Setup	13
5.2.2 WP1: Data Integration	14
5.2.3 WP2: Data Cleaning & Preparation	15
5.2.4 WP3: Exploratory Data Analysis & Feature Engineering	16
5.2.5 WP4: Preprocessing	16
5.2.6 WP5: Model Development (Parallel Tracks)	17

5.2.7	WP6: Model Evaluation & Selection	18
5.2.8	WP7: Deployment & Documentation	19
5.3	Dependencies & Critical Path	20
5.4	Key Milestones	21
5.5	Weekly Status Checklist	21
6	Variable selection and Data Cleaning	22
6.1	Subset the Data	23
6.1.1	Land and Location Attributes	23
6.1.2	Property Characteristics	24
6.1.3	Property Valuation	24
6.1.4	Data Integration	24
6.1.5	Cleaning	25
6.1.6	Data Quality Validation	27
	Appendices	33
A	Appendix: Complete Data Dictionary	33
A.1	Full Field-Level Definitions	33
A.2	Tax Account	33
A.3	Appraisal Account	35
A.4	Improvement	36
A.5	Improvement Built-As	37
A.6	Improvement Detail	39
A.7	Land Attribute	39
A.8	Sale	40
A.9	Seg/Merge	41
A.10	Taxpayer	41
A.11	Tax Description	42
	References	43

Preface

This documentation presents the project proposal and development progress for the Predictive Modeling of Residential Property Prices in Pierce County, developed as part of the Solution Engineering course at FHTW.

The project aims to build a robust machine learning pipeline to estimate housing prices by integrating diverse data sources from the Pierce County Assessor-Treasurer. This book serves as the central repository for project objectives, data descriptions, organizational structure, and development plans.

Project Team

- **Mira Radakovic** (Project Leader)
- **Edberto Moura Lima** (Data Analyst)
- **Gerta Hajrullahi** (Data Analyst)
- **Helga Deutschmann** (Data Analyst)

Document Status

This version corresponds to the **Project Proposal (Report 1)**.

- ☒ Project Objectives (Qualitative & Quantitative)
- ☒ Data Source Description & Dictionary
- ☒ Project Organization & Task Allocation
- ☒ Development Plan & Workflow
- ☒ Gantt Schedule & Milestones

1 Project Objective

1.1 Background & Motivation

Estimating house prices is a classic regression problem with clear real-world importance. Housing markets are influenced by many interconnected factors, including property characteristics, location, neighborhood effects, and market trends over time. Because nearby properties often share similar values and market conditions change continuously, house-price prediction provides an excellent example for applying the complete data-science workflow covered in this course.

For this project, the publicly available Assessor–Treasurer data from Pierce County is used as the main data source. The dataset contains detailed and realistic information about properties, transactions, and assessments across multiple linked tables, offering a large-scale and practical environment for data exploration, pre-processing, feature engineering, and predictive modeling.

1.2 Project Goal

Develop a machine learning system capable of predicting residential property prices using structural, spatial, economic, and neighborhood characteristics.

1.2.1 Qualitative Objectives

1. Build a clean housing dataset for Pierce County.
2. Explore which factors influence house prices.
3. Engineer predictive variables from property and spatial data.
4. Compare traditional statistical and machine learning approaches.
5. Deploy a simple Shiny app for interactive price estimation.
6. Allow non-technical users to test predictions using custom property inputs.
7. Create visual outputs that support decision-making.

1.3 Quantitative Objectives

Table 1.1: Quantitative project success criteria and performance targets.

Objective	Metric	Target
Model Accuracy	Root Mean Square Error (RMSE)	< USD 75,000
Explanatory Power	Coefficient of Determination (R^2)	> 0.85
Model Diversity	Algorithm Benchmarking	4 distinct algorithms
Feature Richness	Engineered Predictors	20 model-ready features
App Performance	Real-time Inference	Response time < 3 seconds
Data Integrity	Table Integration	100% of selected source tables joined

1.4 Deliverables & Timeline

- **21.05.2026:** Project Proposal Presentation
- **06.06.2026:** Interim Results Presentation
- **24.06.2026:** Final Results Presentation & Project Report Submission
- **24.06.2026:** Live demo of deployed model App

2 Data Source Description & Data Dictionary

2.1 Overview

The Pierce County Assessor–Treasurer Data Mart is a comprehensive relational database containing detailed property assessment, taxation, appraisal, land-use, improvement, and sales transaction records maintained by Pierce County, Washington. The dataset integrates information from multiple administrative systems and organizes it into interconnected tables linked primarily through the parcel identification number, enabling consistent tracking of individual properties across taxation, valuation, ownership, and structural characteristics.

The database includes information related to property accounts, land attributes, building improvements, tax assessments, exemption categories, sales history, and geographic identifiers. Together, these tables provide a multidimensional representation of residential and commercial properties and support both fiscal administration and real-estate market analysis.

Here we provide a data dictionary with detailed descriptions of the available tables, attributes, data types, and relationships that form the foundation for exploratory data analysis, spatial analysis, and predictive modeling of residential property prices in Pierce County.

2.2 Relational Diagram

The data mart uses a relational schema with the following structure:

2.3 Data Dictionary Summary

The data mart contains nine related tables. A brief summary follows:

1. **Tax Account:** Current tax and assessment information per parcel
2. **Appraisal Account:** Detailed property appraisal with characteristics

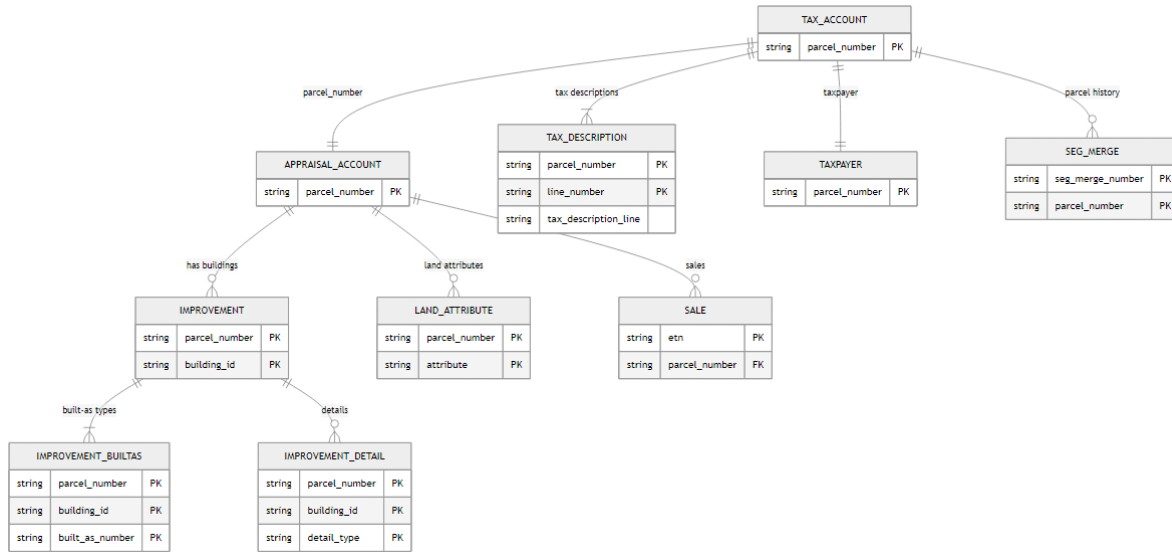


Figure 2.1: Entity–relationship diagram of the Pierce County data showing relationships among tax, appraisal, land, improvement, taxpayer, and sales tables.

3. **Improvement**: Building-level details (multiple per parcel)
4. **Improvement Built-As**: Construction specifications and year built
5. **Improvement Detail**: Features (garages, decks, fireplaces, etc.)
6. **Land Attribute**: Land-specific characteristics affecting valuation
7. **Sale**: Historical property sales transactions
8. **Seg/Merge**: Property segregation and merger history
9. **Taxpayer**: Owner information (privacy restricted)

For detailed field definitions, see the appendix.

3 Project Organization & Task Allocation

3.1 Group Structure

Table 3.1: Project team composition and assigned roles.

Name	Role
Mira Radakovic (PL)	Project Leader
Edberto Moura Lima	Data Analyst
Gerta Hajrullahi	Data Analyst
Helga Deutschmann	Data Analyst

3.2 Tools & Infrastructure

Table 3.2: Technology stack and development infrastructure components.

Component	Detail
Language	R
Database	Supabase (PostgreSQL)
Version Control	Git + GitHub
Project Management	GitHub Projects + conceptboard
Communication	Group messaging + sync meetings
Documentation	Quarto
Data Wrangling	dplyr, tidyr, readr
EDA & Viz	ggplot2, corrplot, leaflet, plotly
Modelling	caret / tidymodels, ranger, xgboost, glmnet
Spatial Analysis	sf, geosphere
Deployment	Shiny, shinydashboard

3.3 Project Kanban Board

The following board tracks the real-time status of all project tasks.



Figure 3.1: Project task tracking board showing workflow stages from backlog to completion.

3.4 Risk Management

Table 3.3: Identified project risks and corresponding mitigation strategies.

Risk	Mitigation
Large number of tables / complex joins	Build join pipeline incrementally; validate row counts at each step
Many missing values in building details	Imputation strategies (median/mode); sensitivity analysis on imputed variables
Data covers multiple property types (commercial, mobile homes)	Filter to Residential Improved sales only at ingestion
Temporal drift in prices (market cycles)	Include sale year/month as features; optionally restrict to recent years
Skewed Sale Price distribution	Log-transform target; evaluate models on both scales
Shiny App complexity	Start with minimal viable app; add features iteratively

4 Development Plan Overview

The project is splinted into work packages (WP0–WP7) over 6 weeks.

Detailed work package specifications (tasks, deliverables, quality criteria, and dependencies) are provided in [Chapter 5: Project Schedule & Work Packages](#).

4.1 Workflow Diagram

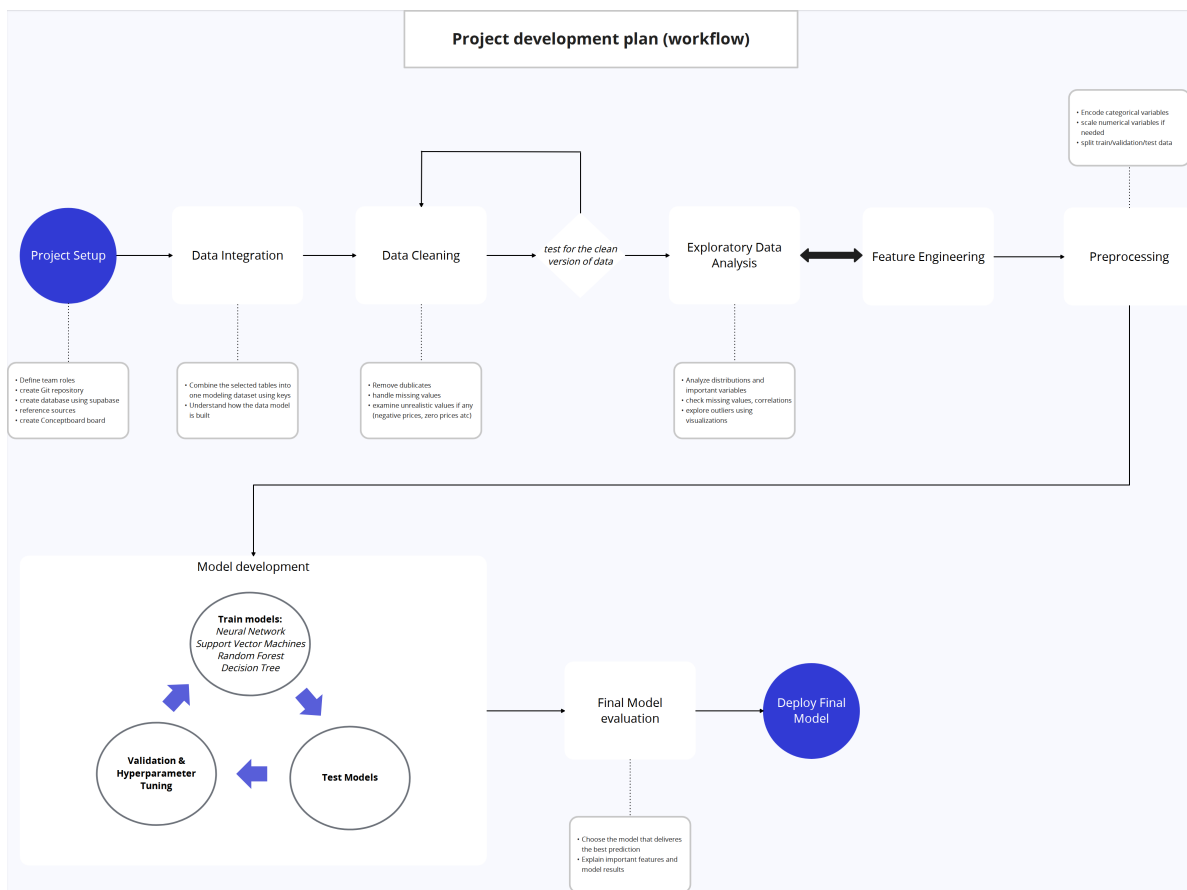


Figure 4.1: Project Development Workflow

4.2 Work Package Summary

Table 4.1: High-level work package phases and planned duration in weeks.

WP	Phase	Duration
WP0	Project Setup	13–19 May
WP1	Data Integration	20–26 May
WP2	Data Cleaning	27 May – 2 June
WP3	EDA & Features	30 May – 6 June
WP4	Preprocessing	7–9 June
WP5	Model Development	9–15 June
WP6	Model Evaluation	16–20 June
WP7	Deployment	20–24 June

4.3 Communication & Oversight

Weekly Sync: - Review completed tasks - Discuss blockers & dependencies - Update Kanban board - Plan next week

Status Reporting: Mira provides weekly status to lecturer

5 Project Schedule & Work Packages

Project Duration: 13 May – 24 June 2026 (6 weeks)

5.1 Interactive Gantt Chart

The following chart outlines the project timeline, milestones, and critical path.

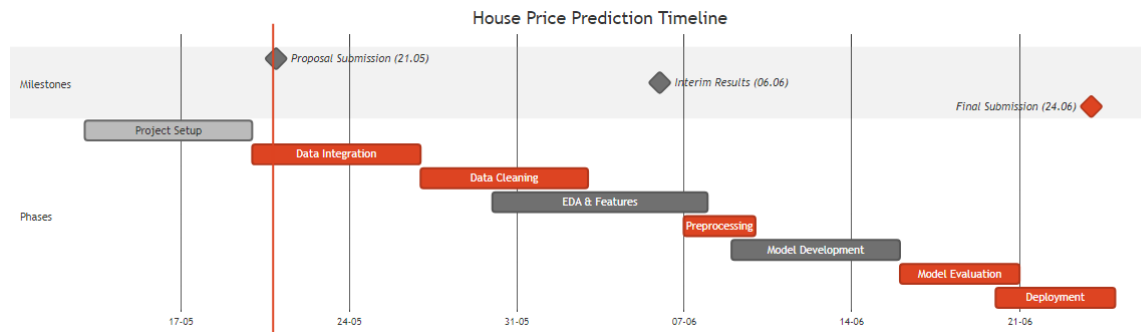


Figure 5.1: Project Gantt Chart showing all work packages and critical path tasks.

5.2 Detailed Work Package Definitions

5.2.1 WP0: Project Setup

Duration: 13–19 May | **Status:** Done

5.2.1.1 Tasks

1. Define team roles and responsibilities
2. Create Git repository (private, shared with lecturer)
3. Set up Supabase database connection
4. Initialize Kanban board (Backlog/To Do/Doing/Testing/Done)
5. Configure R development environment

5.2.1.2 Deliverables

- GitHub repository
 - Supabase connection
 - Kanban board
 - Team charter document
-

5.2.2 WP1: Data Integration

Duration: 20–26 May | **Status:** Partially complete

5.2.2.1 Tasks

1. Download Pierce County data files
2. Map table relationships and join keys
3. Perform sequential table joins (validate at each step)
4. Validate referential integrity and row counts
5. Create data integration report

5.2.2.2 Deliverables

- Integrated dataset (.RData)
- Table mapping diagram
- Data dictionary
- Validation report with row counts

5.2.2.3 Quality Criteria

- ☐ All tables joined successfully
 - ☐ Row counts documented at each step
 - ☐ No unexpected NULL values
 - ☐ Referential integrity verified
-

5.2.3 WP2: Data Cleaning & Preparation

Duration: 27 May – 2 June | **Status:** To Do

5.2.3.1 Tasks

1. Filter to Residential Improved properties only
2. Remove duplicate records
3. Handle missing values (document strategy)
4. Examine and document outliers
5. Variable selection — keep only analysis-ready columns
6. Run validation test suite

5.2.3.2 Deliverables

- Cleaned dataset (.RData)
- Data cleaning log
- Missing value analysis
- Quality assurance report

5.2.3.3 Quality Criteria

- ☐ No critical missing values
 - ☐ Data types verified
 - ☐ Outliers documented
 - ☐ Test suite passes
-

5.2.4 WP3: Exploratory Data Analysis & Feature Engineering

Duration: 30 May – 9 June | **Status:** To Do

5.2.4.1 WP3a: EDA Tasks

- Univariate analysis (distributions, summaries)
- Bivariate analysis (price vs. key variables)
- Spatial analysis with leaflet maps
- Correlation matrix and hypothesis testing

5.2.4.2 WP3b: Feature Engineering Tasks

- Create derived features (age, price/sqft, neighborhood effects, etc.)
- Encode categorical variables
- Feature selection — target 15 final features
- Document feature importance baseline

5.2.4.3 Deliverables

- EDA report with visualizations
- Correlation heatmap
- Feature engineering log
- Final feature set documentation (15 features)

5.2.4.4 Quality Criteria

- ☐ All visualizations publication-ready
 - ☐ 15 features engineered and named
 - ☐ No data leakage from target variable
 - ☐ Feature-target correlations documented
-

5.2.5 WP4: Preprocessing

Duration: 7–9 June | **Status:** To Do

5.2.5.1 Tasks

1. Encode categorical variables (finalize encoding scheme)
2. Scale numerical variables (standardize or normalize)
3. Train/validation/test split (70/15/15% stratified)
4. Document random seed for reproducibility

5.2.5.2 Deliverables

- 3 split datasets (train, validation, test)
- Encoding mapping tables
- Scaling parameters (saved as R object)
- Data split documentation

5.2.5.3 Quality Criteria

- ☐ No data leakage (scaling only from training set)
 - ☐ Balanced distribution across splits
 - ☐ Random seed documented
 - ☐ All features properly encoded/scaled
-

5.2.6 WP5: Model Development (Parallel Tracks)

Duration: 9–15 June | **Status:** To Do

5.2.6.1 WP5a: Baseline & Tree Models

- Baseline Linear Regression
- Decision Tree (hyperparameter tuning)
- Random Forest (grid search over parameters)

5.2.6.2 WP5b: Support Vector Machine

- SVM setup (kernel selection)
- Hyperparameter tuning (C, epsilon)
- Cross-validation

5.2.6.3 WP5c: k-Nearest Neighbours

- Setup (distance metric, k selection)
- kNN based on: distance (lon, lat) + square metre
- Hyperparameter optimization

5.2.6.4 WP5d: Linear Regression

- Full feature set training
- Feature importance extraction
- Residual diagnostics

5.2.6.5 Deliverables

- Trained model objects (.RDS)
- Cross-validation results
- Validation & test metrics
- Model configuration documentation

5.2.6.6 Quality Criteria

- ☐ All 5 models trained and saved
 - ☐ Cross-validation RMSE < 100k (preliminary)
 - ☐ No data leakage to validation/test
 - ☐ Test predictions generated for all samples
-

5.2.7 WP6: Model Evaluation & Selection

Duration: 16–20 June | **Status:** To Do

5.2.7.1 Tasks

1. Cross-validation for all 5 models (k=5 fold)
2. Performance benchmarking (RMSE, MAE, R^2)
3. Create model comparison table and visualizations
4. Select best model based on criteria:
 - $RMSE < USD\ 80,000$
 - $R^2 > 0.80$
5. Extract and visualize feature importance

5.2.7.2 Deliverables

- Model comparison table (CSV)
- Benchmarking plots (PNG/PDF)
- Best model selection report
- Feature importance ranking
- Final model object saved

5.2.7.3 Quality Criteria

- ☐ All models evaluated on same test set
 - ☐ Best model meets $RMSE < 80k$ & $R^2 > 0.80$ targets
 - ☐ Feature importance interpreted
 - ☐ Selection criteria documented
-

5.2.8 WP7: Deployment & Documentation

Duration: 20–24 June | **Status:** To Do

5.2.8.1 Tasks

1. Build Shiny App (UI + server logic)
2. Create prediction endpoint (returns estimate + confidence interval)
3. Add feature importance visualization
4. Final validation testing
5. Prepare presentation slides
6. Assemble final report

7. Deploy to shinyapps.io or local server

5.2.8.2 Deliverables

- Deployed Shiny App
- App source code (app.R)
- Presentation slides (PDF)
- Final report (PDF)
- User guide & model documentation
- All code on GitHub

5.2.8.3 Quality Criteria

- ☐ Shiny App loads without errors
- ☐ Predictions return in <5 seconds
- ☐ Confidence intervals displayed
- ☐ Presentation covers all key findings
- ☐ Report submitted via Moodle by deadline

5.3 Dependencies & Critical Path

Table 5.1: Work package interdependencies and blocking relationships defining the critical path.

WP	Depends On	Notes
WP1	WP0	Requires Git repo & Supabase setup
WP2	WP1	Requires integrated dataset
WP3	WP2	Requires clean dataset
WP4	WP3	Requires feature set
WP5	WP4	Requires preprocessed data
WP6	WP5	Requires all trained models
WP7	WP6	Requires selected best model

Critical Path Tasks: 1. WP1: Data integration (26.05) 2. WP2: Data cleaning (02.06) 3. WP3: Feature engineering (09.06) 4. WP4: Train/test split (09.06) 5. WP5: Model training (15.06) 6. WP6: Model selection (20.06) 7. WP7: Shiny App deployment (23.06)

5.4 Key Milestones

Table 5.2: Project milestones, scheduled events, and corresponding deliverables.

Date	Event	Deliverables
21.05.2026	Proposal Presentation	Objectives, data dictionary, organization, development plan
06.06.2026	Interim Results	Data status, EDA findings, initial model results
18.06.2026	Feedback Round	Review feedback, adjust if needed
24.06.2026	Final Presentation	Live demo + report submission via Moodle

5.5 Weekly Status Checklist

- ☒ **Week 1 (13–19.05):** WP0 setup complete
- ☐ **Week 2 (20–26.05):** WP1 data integration done, Proposal prepared
- ☐ **Week 3 (27.05–02.06):** WP2 data cleaning complete, WP3 started
- ☐ **Week 4 (03–09.06):** WP3 features complete, WP4 preprocessing ready
- ☐ **Week 5 (10–16.06):** WP5 all 5 models trained and validated
- ☐ **Week 6 (17–23.06):** WP6 best model selected, WP7 Shiny App deployed
- ☐ **Week 7 (24.06):** Final presentation & report submitted

6 Variable selection and Data Cleaning

After all the data has been mapped put in a database we construct the final selection of variables used for modelling and clean this data.

```
library(DBI)
library(RPostgres)
```

Warning: package 'RPostgres' was built under R version 4.5.3

```
library(dotenv)
```

Warning: package 'dotenv' was built under R version 4.5.2

```
library(here)
```

here() starts at D:/Documents/00-Personal/01-FHTW/SS2026/MAD-BB-2-SS2026-SOE-R/99-Project-SolutionEngineering/SOE-R-House-Price-Prediction

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

Table 6.1: Selected appraisal account features: land and location attributes

```
df_1 <- dbGetQuery(con, '
  SELECT parcel_no, appraisal_account_type, buildings, land_net_squarefeet, waterfront_type,
  utility_electric, utility_water, street_type, latitude, longitude
  FROM "APPRAISAL_ACCOUNT"
  WHERE appraisal_account_type = \'Residential\'
')
```

```
library(tidyr)
library(forcats)
library(stringr)

load_dot_env(here::here(".env"))
```

```
con <- dbConnect(
  RPostgres::Postgres(),
  host      = Sys.getenv("DB_HOST"),
  port      = as.integer(Sys.getenv("DB_PORT")),
  dbname    = Sys.getenv("DB_NAME"),
  user      = Sys.getenv("DB_USER"),
  password  = Sys.getenv("DB_PASSWORD")
)
dbIsValid(con)
```

```
[1] TRUE
```

6.1 Subset the Data

After mapping the data out and looking at the variable description, we limit the potentially used features. These features should be ones, that a homeowner likely knows about their house.

6.1.1 Land and Location Attributes

Restrict the features to: - land square feet - waterfront type - view quality - electricity availability - water availability - street type - coordinates (longitude, latitude)

Table 6.2: Selected improvement features: building characteristics and construction details

```
df_2 <- dbGetQuery(con, "
  SELECT i.parcel_no, i.building_id, i.property_type, i.condition,
         b.stories, b.bedrooms, b.bathrooms, b.units, b.year_built, b.year_remodeled
  FROM \"IMPROVEMENT\" i
  LEFT JOIN \"IMPROVEMENT_BUILT_AS\" b
    ON i.parcel_no = b.parcel_no
    AND i.building_id = b.building_id
  WHERE i.property_type = 'Residential'
")
```

Table 6.3: Market value from tax account table (target variable)

```
df_3 <- dbGetQuery(con, '
  SELECT parcel_no, total_market_value_current_year
  FROM "TAX_ACCOUNT"
')
```

6.1.2 Property Characteristics

We get property specifics: - condition - number of stories - number of bedrooms - number of bathrooms - units - year built - year remodeled

6.1.3 Property Valuation

We get the current property value:

6.1.4 Data Integration

We join the data together from all three source subsets:

Table 6.4: Integrated dataset combining land, improvement, and valuation features

```
df_all <- df_1 %>%
  left_join(df_2, by = "parcel_no") %>%
  left_join(df_3, by = "parcel_no")
```


Table 6.5: Initial assessment of missing values across all features

```
colSums(is.na(df_all))
```

parcel_no	appraisal_account_type
0	0
buildings	land_net_squarefeet
0	3
waterfront_type	view_quality
256540	252509
utility_electric	utility_water
0	0
street_type	latitude
0	5
longitude	building_id
5	29816
condition	stories
29816	29846
bedrooms	bathrooms
29846	29846
units	year_built
29887	29846
year_remodeled	total_market_value_current_year
42869	0

6.1.5 Cleaning

We obtain the dataset with selected features. Now we go ahead to prepare the data for initial exploratory data analysis.

```
df_all <- df_all %>%
  select( -property_type)
```

We first select lands with buildings on it, as we are interested in residential home valuations.

```
df_all <- df_all %>%
  filter(buildings != 0)
```

Then we look at the missing values and treat them one-by-one.

We remove observations that have no bedroom or bathroom recorded as these would be one of the features easiest for the user to input and we want to use them. If they are missing it likely hints at a data recording issue or a building type that is not a classical residential home (which is out of scope for this project).

```
df_all <- df_all %>%
  filter(!is.na(bathrooms))
```

We also remove the 5 observations missing coordinates or square footage.

```
df_all <- df_all %>%
  filter(!is.na(latitude) & !is.na(longitude) & !is.na(land_net_squarefeet))
```

Waterfront type records River, Lake, Stream/Creek or Salt. We fill in missing values to represent no waterfront, as such category does not exist and missing specification likely represents that the property has no direct water connection.

```
df_all <- df_all %>%
  mutate(waterfront_type = replace_na(waterfront_type, "None"))
unique(df_all$waterfront_type)
```

```
[1] "WF Salt"          "None"             "WF Lake"          "WF Stream/Creek"
[5] "WF River"
```

View quality currently has 9 categories that is slightly excessive, and more confusing for the potential user and increases model complexity.

```
unique(df_all$view_quality)
```

```
[1] NA                "View Lim"        "View Lim -"     "View Lim +"
[5] "View Avg +"      "View Avg"        "View Good"      "View Good +"
[9] "View V-Good"     "View V-Good +"
```

We collapse the categories into “limited”, “average”, “good”, “very good”. We replace NAs with uncategorized.

```
df_all <- df_all %>%
  mutate(
    view_quality = replace_na(view_quality, "uncategorized"),

    view_quality = fct_collapse(view_quality,
      "limited" = c("View Lim", "View Lim -", "View Lim +"),
      "average" = c("View Avg", "View Avg +"),
      "good" = c("View Good", "View Good +"),
      "very good" = c("View V-Good", "View V-Good +")
    )
  )
unique(df_all$view_quality)
```

```
[1] uncategorized limited      average      good          very good
Levels: uncategorized average good limited very good
```

We order the view types.

```
df_all$view_quality <- factor(df_all$view_quality,
                             levels = c("uncategorized", "limited", "average", "good", "very good"),
                             ordered = TRUE)
```

For the year remodeled, we will proceed as no remodeling happened, if the year is not recorded.

```
df_all <- df_all %>%
  mutate(year_remodeled = replace_na(year_remodeled, 0))
```

Lastly, in the units feature we convert the few missing values to 1 unit after looking into the data manually, and also convert 0 units to 1 as the empirical meaning of having no separate units and the whole unit being one (for example family home) is the same.

```
df_all <- df_all %>%
  mutate(units = if_else(is.na(units) | units == 0, 1, units))
```

We have now the complete dataset with the preselected features.

6.1.6 Data Quality Validation

A few post checks we should still apply.

Table 6.6: Final verification of missing values after cleaning (all zeros indicate complete data)

```
colSums(is.na(df_all))
```

parcel_no	appraisal_account_type
0	0
buildings	land_net_squarefeet
0	0
waterfront_type	view_quality
0	0
utility_electric	utility_water
0	0
street_type	latitude
0	0
longitude	building_id
0	0
condition	stories
0	0
bedrooms	bathrooms
0	0
units	year_built
0	0
year_remodeled	total_market_value_current_year
0	0

```
unique(df_all$utility_electric)
```

```
[1] "POWER INSTALLED"      "POWER AVAILABLE"      "POWER NO - COMMENT"
```

```
df_all <- df_all %>%
  mutate(utility_electric = str_replace(utility_electric, "POWER NO - COMMENT", "POWER NO"))
unique(df_all$utility_electric)
```

```
[1] "POWER INSTALLED" "POWER AVAILABLE" "POWER NO"
```

```
unique(df_all$utility_water)
```

```
[1] "WATER INSTALLED" "WATER NO"          "WATER AVAILABLE"
```

```
unique(df_all$street_type)
```

```
[1] "STREET UNPAVED" "PAVED"             "STREET NO ROAD"
```

Check coordinate range.

```
df_all %>%
  summarise(
    min_lat = min(latitude, na.rm = TRUE),
    max_lat = max(latitude, na.rm = TRUE),
    min_lon = min(longitude, na.rm = TRUE),
    max_lon = max(longitude, na.rm = TRUE)
  )
```

Table 6.7: Geographic bounding box of properties in the dataset

min_lat	max_lat	min_lon	max_lon
46.73973	47.40335	-122.8406	-121.4681

```
unique(df_all$condition)
```

```
[1] "Average"      "Extra Poor"    "Uninhabitable" "Poor"
[5] "Fair"         "Very Poor"
```

```
df_all$condition <- factor(df_all$condition,
                           levels = c("Uninhabitable",
                                       "Extra Poor",
                                       "Very Poor",
                                       "Poor",
                                       "Fair",
                                       "Average"),
                           ordered = TRUE)
```

```
df_all %>%
  select(stories, bedrooms, bathrooms) %>%
  slice_max(order_by = stories, n = 5)
```

stories	bedrooms	bathrooms
15	3	2.00
10	3	2.00
8	0	0.50
4	3	3.75
4	3	4.00

The top 3 observation are very unlikely with this combination of bedrooms and bathrooms, so we remove them.

```
to_remove <- df_all %>%
  slice_max(order_by = stories, n = 3, with_ties = FALSE)

df_all <- df_all %>%
  anti_join(to_remove)
```

Joining with `by = join_by(parcel_no, appraisal_account_type, buildings, land_net_squarefeet, waterfront_type, view_quality, utility_electric, utility_water, street_type, latitude, longitude, building_id, condition, stories, bedrooms, bathrooms, units, year_built, year_remodeled, total_market_value_current_year)`

Looking at the data again manually, and in particular the combination of bathrooms and bedrooms with teh units we also detect some anomalies that we remove.

```
df_all <- df_all %>%
  filter(bathrooms <= 17 & bedrooms <= 21)
```

The year remodeled being 0 means no remodeling has been recorded on the house.

```
df_all %>%
  summarise(
    min_built      = min(year_built, na.rm = TRUE),
    max_built      = max(year_built, na.rm = TRUE),
    min_remodeled  = min(year_remodeled, na.rm = TRUE),
    max_remodeled  = max(year_remodeled, na.rm = TRUE)
  )
```

Table 6.8: Range of construction and renovation years (0 = no remodeling recorded)

min_built	max_built	min_remodeled	max_remodeled
1861	2026	0	2026

We additionally check if the year remodeled is not 0 it is always more recent than the year built.

```
impossible_dates <- df_all %>%
  filter(year_remodeled > 0 & year_remodeled < year_built)

nrow(impossible_dates)
```

```
[1] 0
```

Check the valuation range.

```
df_all %>%
  summarise(
    min_value      = min(total_market_value_current_year, na.rm = TRUE),
    max_value      = max(total_market_value_current_year, na.rm = TRUE)
  )
```

Table 6.9: Property valuation range in current year market values (in USD)

min_value	max_value
8400	10627600

```
#save(df_all, file = here::here("data", "df_all.RData"))
```


A Appendix: Complete Data Dictionary

A.1 Full Field-Level Definitions

This appendix contains the complete field-level data dictionary for all nine tables in the Pierce County Assessor-Treasurer Data Mart. These detailed descriptions should be consulted as reference during data integration, feature engineering, and validation.

Source: Pierce County Assessor-Treasurer Office <https://online.co.pierce.wa.us/cfapps/atr/data-mart/metadata/>

A.2 Tax Account

Comprehensive tax and assessment account information for each property.

Table A.1: Tax Account table - tax and assessment account information.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Account Type	Text	5	REAL - Real Property; STRUC - Structures; PERS - Personal Property; MOBIL - Mobile Home
Property Type	Text	5	ASIMP - Administrative Seg Improvement; LNDIM - Land and Improvements; MBLHM - Mobile Home; RRLEA - Railroad Leases; SAPP - State Assessed Personal Property; SARP - State Assessed Real Property; STRUC - Leased Land and/or Structure
Site Address	Text	50	Physical location address of the property. May be different than mailing address. "XXX" indicates estimated address using GIS

Field Name	Type	Size	Description
Use Code	Text	5	4-digit numeric code for present highest and best use of the property
Use Description	Text	50	Text description of the numeric use code
Tax Year - Prior	Integer		Tax year for which all prior year values apply
Tax Code Area - Prior Year	Text	10	Code describing geographical area and specific combination of taxing districts. Subject to annual change
Exemption Type - Prior Year	Text	40	Type of exemption, if any, applied to the property
Current Use Code - Prior Year	Text	5	AGRI - Agricultural; CT - Current Use Timber; FORDG - Forest Designated; OPBRS - Open Space PBRS; OPEN - Open Space
Land Value - Prior Year	Integer		Total fair market value of land, as determined by appraisal process
Improvement Value - Prior Year	Integer		Total fair market value of improvements (buildings, structures), as determined by appraisal process
Total Market Value - Prior Year	Integer		Total fair market value of land and improvements, as determined by appraisal process
Taxable Value - Prior Year	Integer		Market value of the property, minus any exemptions granted
Tax Year - Current	Integer		Tax year for which all current year values apply
Tax Code Area - Current Year	Text	10	Code describing geographical area and specific combination of taxing districts
Exemption Type - Current Year	Text	40	Type of exemption, if any, applied to the property
Current Use Code - Current Year	Text	5	AGRI - Agricultural; CT - Current Use Timber; FORDG - Forest Designated; OPBRS - Open Space PBRS; OPEN - Open Space
Land Value - Current Year	Integer		Total fair market value of land for the parcel
Improvement Value - Current Year	Integer		Total fair market value of improvements (buildings, structures) for the parcel
Total Market Value - Current Year	Integer		Total fair market value of land and improvements for the parcel

Field Name	Type	Size	Description
Taxable Value - Current Year	Integer		Market value of the property, minus any exemptions granted
Range	Text	10	Range
Township	Text	10	Township
Section	Text	20	Section
Quarter Section	Text	20	Quarter
Subdivision Name	Text	50	Subdivision name
Located On Parcel	Text	10	Real property parcel number on which this improvement-only parcel is located (Personal Property, Mobile Homes, Leasehold Structures)

A.3 Appraisal Account

Detailed property appraisal data with characteristics and valuations.

Table A.2: Appraisal Account table - property appraisal data with characteristics.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Appraisal Account Type	Text	15	Com Condo, Com Leasehold, Com Multi Unit, Commercial, Industrial, Mobile Home, Reference, Res Com Condo, Res Leasehold, Residential, Trended Invest
Business Name	Text	50	Main business name on the parcel, or space number for mobile/manufactured home in a park
Value Area ID	Text	10	Identifies the Physical Inspection cycle for each parcel (PI1 to PI6)
Land Economic Area	Text	30	Ties parcels to land valuation models; identifies land use model by PI year or appraisal area/sector/subsector
Buildings	Integer		Represents the number of improvements (buildings) located on a parcel
Group Account Number	Text	30	Identifies all parcels as one economic unit or condominium project

Field Name	Type	Size	Description
Land Gross Acres	Decimal	15.4	Total acres that make up the economic unit which could include multiple parcels
Land Net Acres	Decimal	15.4	Size of the individual parcel in acres
Land Gross Square Feet	Decimal	15.4	Total square feet that make up the economic unit which could include multiple parcels
Land Net Square Feet	Decimal	15.4	Size of the individual parcel in square feet
Land Gross Front Feet	Decimal	15.4	Front feet of site facing arterial (commercial), address side (residential), or waterfront length
Land Width	Integer		Net Effective Waterfront Front Feet of individual parcel (Residential Use Only)
Land Depth	Integer		Net Effective Depth from top of bank to inland property line (Residential Use Only)
Submerged Area Square Feet	Integer		Area of a parcel that is submerged; not included in Gross or Net Square Feet
Appraisal Date	Date		Most recent date an appraiser physically observed the property and/or made changes
Waterfront Type	Text	30	Describes the type of waterfront the property adjoins or has legal access to
View Quality	Text	30	Market appeal rating of overall view available from dwelling or property
Utility Electric	Text	30	Identifies if power is installed, available, or not available on the property
Utility Sewer	Text	30	Identifies if sewer/septic is installed, available, or not available
Utility Water	Text	30	Identifies if water is installed, available, or not available
Street Type	Text	30	Identifies if access street is paved or unpaved
Latitude	Decimal	10.5	Latitude of parcel centroid (some parcels may not have coordinates)
Longitude	Decimal	10.5	Longitude of parcel centroid

A.4 Improvement

Detailed building improvement data linked to appraisal accounts.

Table A.3: Improvement table - building improvement data linked to appraisal accounts.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel
Property Type	Text	15	Links buildings to cost and depreciation tables: Commercial, Duplex, Industrial, Mobile Home, Multiple Unit, Out Building, Residential, Townhouse, Triplex
Neighborhood	Text	10	Boundaries for land subject to similar social, environmental, economic, and governmental forces
Neighborhood Extension	Text	10	Based on property type: primary occupancy (Commercial), zero (Residential), or park designation (Mobile Homes)
Square Feet	Integer		Sum of square feet for 'built as' types (may represent count for Storage Tank types)
Net Square Feet	Integer		Sum of net rentable area(s) for commercial; equal to built-as square feet or zero for residential

A.5 Improvement Built-As

Detailed specifications for building construction styles and features.

Table A.4: Improvement Built-As table - building construction styles and features.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel
Built-As Number	Integer		Sequential number starting with largest Built-As being number 1
Built-As ID	Integer		Number associated with the description of the purpose or style of construction

Field Name	Type	Size	Description
Built-As Description	Text	255	Original purpose/style of construction; associated with Marshall and Swift cost and depreciation tables
Built-As Square Feet	Integer		Sum of square feet for the 'built as' types identified for the building
HVAC	Integer		Code associated with the predominant heating source for the built-as structure
HVAC Description	Text	30	Text description of heating source (Forced Air, Electric Baseboard, Steam, etc)
Exterior	Text	25	Predominant type of construction materials used for exterior siding (Residential Buildings)
Interior	Text	15	Predominant type of interior wall materials (Sheetrock or Paneling; Residential and Mobile Homes only)
Stories	Decimal	13.2	Number of floors/building levels above grade (excluding attic or basement)
Story Height	Integer		Average ceiling heights (Commercial) or majority story height for first floor (Residential)
Sprinkler Square Feet	Integer		Total square footage covered by a sprinkler system
Roof Cover	Text	20	Material used for the roof (Composition Shingles, Wood Shake, Concrete Tile, etc)
Bedrooms	Integer		Number of bedrooms listed for residential property
Bathrooms	Decimal	7.2	Number of baths listed as decimal (e.g., 2.75 = two full + one three-quarter)
Units	Integer		Number of separate units within the building
Class Code	Text	20	Code for one of five basic cost groups by type of framing, walls, floors, roof, and fireproofing
Class Description	Text	50	Text description for cost group classification
Year Built	Integer		Year the building was built per building permit or historical record
Year Remodeled	Integer		Year improvements were made extending the life of the structure
Adjusted Year Built	Integer		Greater than Year Built if improvements reduce the effective age of the building
Physical Age	Integer		Tax year less adjusted year built

Field Name	Type	Size	Description
Built-As Length	Integer		Length of the mobile/manufactured home
Built-As Width	Integer		Width of the mobile/manufactured home
Mobile Home Model	Text	50	Model of the mobile/manufactured home

A.6 Improvement Detail

Additional specific improvement characteristics and features.

Table A.5: Improvement Detail table - building characteristics and features.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel
Detail Type	Text	10	Category: Add On, Appliance, Balcony, Basement, Carport, Elevator, Fixture, Garage, Mezzanine, Porch, Rough In, Storage
Detail Description	Text	50	Brief explanation of the improvement characteristic (Fireplace, Laundry Facility, Type of Porch, Type of Bath, etc)
Units	Decimal	15.4	Component parts indicating either square footage of the structure or count of each component

A.7 Land Attribute

Land-specific attributes and characteristics affecting valuation.

Table A.6: Land Attribute table - land-specific characteristics affecting valuation.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Attribute	Text	30	Category: Residential (R), Commercial (C), or Res Com Condo (X) with subcategory
Attribute Description	Text	30	Further defines the land attribute (View Avg, NBHD Gated, WF Bank Low, water available, etc)

A.8 Sale

Historical property sales transaction data.

Table A.7: Sale table - historical property sales transaction data.

Field Name	Type	Size	Description
ETN	Text	11	Recording number issued by the Auditor's Office when property transaction is recorded
Parcel Count	Integer		Number of parcels associated with a sale
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Sale Date	Date		Date the legal document (deed) was executed
Sale Price	Decimal	15.2	Dollar amount recorded on the ETN
Deed Type	Text	40	Type of document conveying interest in or legal title to the property
Grantor	Text	80	Individual(s) or company conveying ownership or interest in the property
Grantee	Text	80	Individual(s) or company purchasing ownership or interest in the property
Valid/Invalid	Text	7	Code for validity of a sale for appraisal purposes per State Ratio RCW
Confirmed/Unconfirmed	Text	11	Whether property characteristics and sale circumstances were verified by Assessor-Treasurer staff

Field Name	Type	Size	Description
Exclude Reason	Text	30	State required description of why a sales transaction must be excluded from sales studies
Improved/Vacant	Text	8	Coded as improved if any parcel has building value; vacant for land with no building value
Appraisal Account Type	Text	15	Parcel classification (Com Condo, Com Leasehold, Commercial, Industrial, Residential, etc)

A.9 Seg/Merge

Property segregation and merger transaction history.

Table A.8: Seg/Merge table - property segregation and merger transaction history.

Field Name	Type	Size	Description
Seg/Merge Number	Text	12	Unique number assigned to each segregation
Parent/Child Indicator	Text	1	P = Parent Parcel (inactivated); C = Child Parcel (newly created from segregation)
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Continued Indicator	Text	1	Y = parcel still active after segregation; N = parcel no longer active
Completed Date	Date		Date that the segregation was completed
Tax Year	Text	4	Tax year of the segregation

A.10 Taxpayer

Property owner information (subject to privacy restrictions).

Table A.9: Taxpayer table - property owner information (privacy restricted).

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property

A.11 Tax Description

Detailed tax coding and descriptions for parcels.

Table A.10: Tax Description table - tax coding and descriptions for parcels.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Line Number	Integer		Sequential number identifying each line of a tax description
Tax Description Line	Text	200	Tax description text

References